

Multi-Zone VAV Building Model — Complete Overview

File: `MultiZone_VAV_PythonControl.idf` **EnergyPlus Version:** 25.2 **Purpose:** Control-free plant for runtime Python control — no EMS or hardcoded control logic. All control is driven externally via the EnergyPlus Python API.

1. Site & Climate

Parameter	Value
Location	Colombo, Sri Lanka
Latitude	6.90° N
Longitude	79.86° E
Time Zone	UTC +5.5
Elevation	7.0 m
Climate Type	Hot & humid tropical
Weather File Required	Colombo <code>.epw</code>
Terrain	City

Design Days

Design Day	Month/Day	Dry-Bulb (°C)	Daily Range (°C)	Wet-Bulb (°C)	Wind (m/s)	Pressure (Pa)	Sky Clearness
Cooling (Summer)	Apr 21	33.0	7.0	27.0	3.0	101,200	1.0 (Clear)
Heating (Winter)	Jan 21	22.0	0.0	21.0	3.0	101,200	0.0 (Overcast)

Simulation Settings

Parameter	Value
Timestep	6 per hour (10-minute intervals)
Run Period	Jan 1 - Dec 31 (full year)
Start Day	Monday
Solar Distribution	FullExterior
Sizing Safety Factor	1.2 (both heating and cooling)
Zone Sizing Calculation	Yes
System Sizing Calculation	Yes
Run for Weather File	Yes
Run for Sizing Periods	No

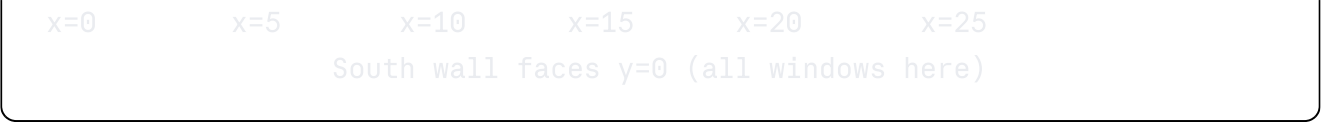
2. Building Geometry

The building is a single-storey, rectangular structure measuring **25 m × 8 m × 3 m** (L × W × H), divided into **5 zones** arranged side-by-side along the east-west axis. Each zone is 5 m wide and 8 m deep with a 3 m ceiling height.

Dimension	Value
Total Footprint	25 m × 8 m = 200 m²
Ceiling Height	3.0 m
Total Volume	600 m³
Number of Zones	5
Zone Footprint (each)	5 m × 8 m = 40 m²
Zone Volume (each)	120 m³

Zone Layout (West → East)





Wall Boundary Conditions per Zone

Every zone has South and North walls exposed to **Outdoors** (SunExposed, WindExposed). The inter-zone partition walls (East/West) are treated as **Adiabatic** interior walls. Zone 1's West wall and Zone 5's East wall are exterior.

Zone	South	North	West	East	Roof	Floor
Zone 1	Exterior	Exterior	Exterior	Adiabatic	Exterior	Adiabatic
Zone 2	Exterior	Exterior	Adiabatic	Adiabatic	Exterior	Adiabatic
Zone 3	Exterior	Exterior	Adiabatic	Adiabatic	Exterior	Adiabatic
Zone 4	Exterior	Exterior	Adiabatic	Adiabatic	Exterior	Adiabatic
Zone 5	Exterior	Exterior	Adiabatic	Exterior	Exterior	Adiabatic

Windows (Fenestration)

Each zone has one south-facing window. All windows are 3 m wide × 1.5 m tall, positioned between 1.0 m and 2.5 m above the floor.

Zone	Window	Width × Height	Area	WWR (South wall)
Zone 1	Zone 1 Window South	3.0 m × 1.5 m	4.5 m ²	30%
Zone 2	Zone 2 Window South	3.0 m × 1.5 m	4.5 m ²	30%
Zone 3	Zone 3 Window South	3.0 m × 1.5 m	4.5 m ²	30%
Zone 4	Zone 4 Window South	3.0 m × 1.5 m	4.5 m ²	30%
Zone 5	Zone 5 Window South	3.0 m × 1.5 m	4.5 m ²	30%

3. Materials & Constructions

Materials

Name	Roughness	Thickness (m)	Conductivity (W/m·K)	Density (kg/m ³)	Specific Heat (J/kg·K)	Thermal Absorpta
M_Stucco	MediumRough	0.025	0.72	1,856	840	0.90
M_Concrete200	MediumRough	0.200	1.95	2,240	900	0.90
M_Insulation	MediumRough	0.075	0.049	265	836	0.90
M_Gypsum	Smooth	0.016	0.16	800	1,090	0.90
M_RoofMembrane	VeryRough	0.010	0.16	1,120	1,460	0.90
M_RoofInsulation	MediumRough	0.120	0.049	265	836	0.90

Name	Roughness	Thickness (m)	Conductivity (W/m·K)	Density (kg/m³)	Specific Heat (J/kg·K)	Thermal Absorpta
M_FloorSlab	MediumRough	0.100	1.31	2,240	900	0.90

Glazing

Property	Value
Type	SimpleGlazingSystem
U-Factor	3.0 W/m²·K
SHGC	0.40
Visible Transmittance	0.60

Construction Assemblies (Outside → Inside)

Construction	Layers	Total Thickness
Exterior Wall	Stucco → Concrete200 → Insulation → Gypsum	0.316 m
Interior Wall	Gypsum → Gypsum	0.032 m
Roof	RoofMembrane → RoofInsulation → Concrete200	0.330 m
Floor	FloorSlab	0.100 m
Exterior Window	SimpleGlazing	—

4. Thermal Zones & Internal Loads

Zone Summary

Property	Zone 1: Open Office	Zone 2: Private Offices	Zone 3: Conference Room	Zone 4: Server Room	Zone 5: Reception
Floor Area	40 m ²	40 m ²	40 m ²	40 m ²	40 m ²
Volume	120 m ³	120 m ³	120 m ³	120 m ³	120 m ³
Occupancy	8 people	4 people	12 people	1 person	5 people
Occ. Density	0.20 p/m ²	0.10 p/m ²	0.30 p/m ²	0.025 p/m ²	0.125 p/m ²
Activity Level	120 W/person	120 W/person	120 W/person	120 W/person	120 W/person
CO ₂ Generation	3.82×10 ⁻⁸ m ³ /s·W	3.82×10 ⁻⁸ m ³ /s·W	3.82×10 ⁻⁸ m ³ /s·W	3.82×10 ⁻⁸ m ³ /s·W	3.82×10 ⁻⁸ m ³ /s·W
Lighting	10 W/m ²	9 W/m ²	8 W/m ²	6 W/m ²	11 W/m ²
Light Return Air	0.42	0.42	0.42	0.42	0.42
Light Radiant	0.18	0.18	0.18	0.18	0.18
Equipment	12 W/m ²	10 W/m ²	5 W/m ²	60 W/m ²	6 W/m ²

Property	Zone 1: Open Office	Zone 2: Private Offices	Zone 3: Conference Room	Zone 4: Server Room	Zone 5: Reception
Equip. Schedule	Office Equipment	Office Equipment	Office Equipment	Server (0.90 const)	Office Equipment
Equip. Radiant	0.30	0.30	0.30	0.30	0.30
Infiltration	0.0003 m ³ /s·m ²	0.0003 m ³ /s·m ²	0.0003 m ³ /s·m ²	0.0003 m ³ /s·m ²	0.0003 m ³ /s·m ²
OA Requirement	0.008 m ³ /s·p	0.008 m ³ /s·p	0.010 m ³ /s·p	0.005 m ³ /s·p	0.010 m ³ /s·p
Cooling Setpoint	24 °C	24 °C	23 °C	21 °C	24 °C
Heating Setpoint	18 °C	18 °C	18 °C	18 °C	18 °C
Thermostat Type	DualSetpoint	DualSetpoint	DualSetpoint	DualSetpoint	DualSetpoint
Control Type	4 (Dual)	4 (Dual)	4 (Dual)	4 (Dual)	4 (Dual)

Key observations:

- Zone 4 (Server Room) has the highest equipment load at 60 W/m² (nearly constant at 90% schedule), the tightest cooling setpoint at 21 °C, and only 1 occupant — this is the most cooling-intensive zone.

- Zone 3 (Conference Room) has the highest occupancy density at 12 people in 40 m² and a higher OA requirement per person (0.01 m³/s).
- Total peak occupancy across all zones: 30 people.

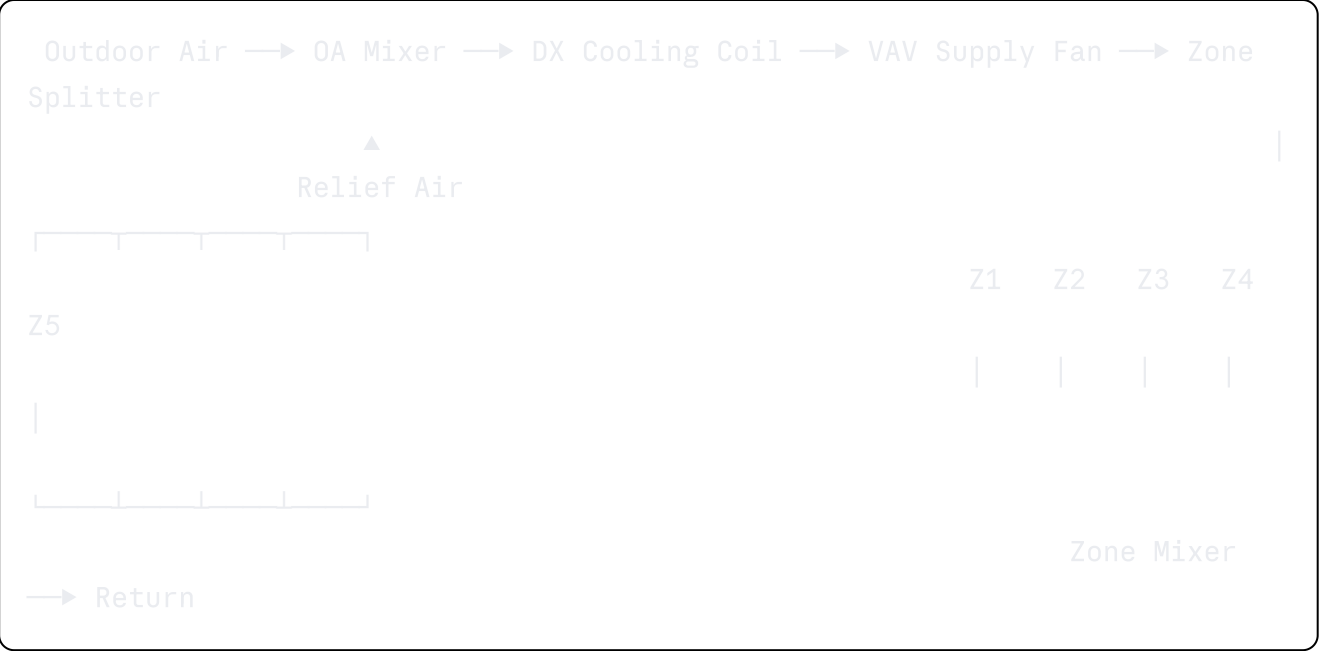
Schedules

Schedule	Type	Weekday Profile	Weekend/Holiday
Office Occupancy	Fraction	0.05 (00–08h), 1.00 (08–18h), 0.05 (18–24h)	0.0 all day
Office Lighting	Fraction	0.10 (00–08h), 0.90 (08–18h), 0.10 (18–24h)	0.05 all day
Office Equipment	Fraction	0.30 (00–08h), 0.90 (08–18h), 0.30 (18–24h)	0.20 all day
Server Equipment	Fraction	0.90 constant (24/7)	0.90 constant
Always On	Fraction	1.0 constant	1.0 constant

5. HVAC System

System Overview

The building is served by a **single central Variable Air Volume (VAV) Air Handling Unit** with DX cooling. There is no heating coil — appropriate for Colombo's tropical climate.



AHU Components

Component	Type	Key Specifications
Air Loop	AirLoopHVAC	"VAV Air Loop", autosize design flow
Outdoor Air System	Controller:OutdoorAir	No economizer, ModulateFlow, FixedMinimum
OA Mixer	OutdoorAir:Mixer	Mixed air at OA Mixer Outlet Node
Cooling Coil	Coil:Cooling:DX:SingleSpeed	COP = 3.5, autosize capacity/SHR/airflow
Coil System	CoilSystem:Cooling:DX	Sensible control, no dehumidification control
Supply Fan	Fan:VariableVolume	$\eta = 0.70$, $\Delta P = 600$ Pa, motor $\eta = 0.90$, min flow = 25%

Component	Type	Key Specifications
SAT Setpoint	SetpointManager:Scheduled	Default 13 °C (Python overrides at runtime)

DX Cooling Coil Performance Curves

Curve	Type	Purpose
DX CapFT	Biquadratic	Total capacity as f(Twb,entering, Tdb,outdoor)
DX CapFFF	Quadratic	Capacity correction for flow fraction
DX EIRFT	Biquadratic	EIR as f(Twb,entering, Tdb,outdoor)
DX EIRFFF	Quadratic	EIR correction for flow fraction
DX PLF	Quadratic	Part-load fraction (degradation), min PLF = 0.85

Fan Power Coefficients

The VAV fan uses a 5th-order polynomial for part-load power: coefficients are [0.0407, 0.08804, -0.07292, 0.9433, 0.0].

Zone Air Terminals

All 5 zones use `AirTerminal:SingleDuct:ConstantVolume:NoReheat` terminals. EnergyPlus does not modulate these internally — the Python controller overrides each terminal's mass flow rate every timestep.

Zone	Terminal Name	Max Airflow (m³/s)
Zone 1 — Open Office	Zone 1 Air Terminal	0.50
Zone 2 — Private Offices	Zone 2 Air Terminal	0.35
Zone 3 — Conference Room	Zone 3 Air Terminal	0.55
Zone 4 — Server Room	Zone 4 Air Terminal	0.60
Zone 5 — Reception	Zone 5 Air Terminal	0.35
Total		2.35

System Sizing Parameters

Parameter	Value
Cooling Design SAT	13.0 °C
Heating Design SAT	40.0 °C
Cooling Design Supply Air Humidity Ratio	0.0085 kg/kg
Heating Design Supply Air Humidity Ratio	0.008 kg/kg
Preheat Temperature	7.0 °C
Zone Sum Method	NonCoincident
System OA Method	ZoneSum
Max OA Fraction	1.0

Parameter	Value
Cooling/Heating Capacity Method	DesignCapacity, autosize

6. CO₂ Contaminant Balance

Parameter	Value
CO ₂ Simulation	Enabled
Outdoor CO ₂ Concentration	400 ppm (constant)
Generic Contaminant	Disabled
CO ₂ Generation per Person	3.82×10^{-8} m ³ /s per W of activity

This enables tracking of indoor CO₂ levels for demand-controlled ventilation strategies via the Python controller.

7. Python API Control Points

The model is designed as a "blank canvas" — all control logic is external. The IDF provides initial default setpoints for warm-up and sizing, which the Python controller overrides at each timestep.

Available Actuators

Control Target	Actuator Component	Control Type	Description
Central SAT	DX Coil Outlet Node	System Node Setpoint (Temp)	Supply air temperature setpoint
Zone 1 Airflow	Zone 1 Air Terminal	Mass Flow Rate (kg/s)	Per-zone supply air flow
Zone 2 Airflow	Zone 2 Air Terminal	Mass Flow Rate (kg/s)	Per-zone supply air flow
Zone 3 Airflow	Zone 3 Air Terminal	Mass Flow Rate (kg/s)	Per-zone supply air flow
Zone 4 Airflow	Zone 4 Air Terminal	Mass Flow Rate (kg/s)	Per-zone supply air flow
Zone 5 Airflow	Zone 5 Air Terminal	Mass Flow Rate (kg/s)	Per-zone supply air flow
Zone Setpoints	Zone Temperature Control	Heating/Cooling SP	Per-zone thermostat setpoints
Outdoor Air	Outdoor Air Controller	Air Mass Flow Rate	OA damper flow control

Key Output Variables (at Timestep resolution)

Category	Variables
Zone State	Mean Air Temperature, Humidity Ratio, Relative Humidity, CO ₂ Concentration
Thermostat	Cooling Setpoint Temperature, Heating Setpoint Temperature
Node Data	Mass Flow Rate, Temperature, Humidity Ratio, Relative Humidity, Setpoint Temperature, CO ₂ Concentration
Outdoor	Site Outdoor Air Drybulb Temperature, Humidity Ratio, Air System OA Mass Flow Rate
Energy	Fan Electricity Rate, Cooling Coil Electricity Rate, Total/Sensible/Latent Cooling Rate

8. Summary at a Glance

Attribute	Value
Building Type	Commercial office (single storey)
Location	Colombo, Sri Lanka (tropical)
Total Floor Area	200 m ²
Zones	5 (Open Office, Private Offices, Conference, Server Room, Reception)
Total Peak Occupancy	30 people

Attribute	Value
HVAC	Central VAV with DX cooling, no heating coil
Cooling COP	3.5
Fan	Variable volume, 600 Pa, 70% efficiency
Default SAT	13 °C
Cooling Setpoints	21-24 °C (zone-dependent)
Heating Setpoint	18 °C (all zones)
CO ₂ Tracking	Enabled (400 ppm outdoor baseline)
Control Philosophy	Fully external via EnergyPlus Python API
Simulation Timestep	10 minutes